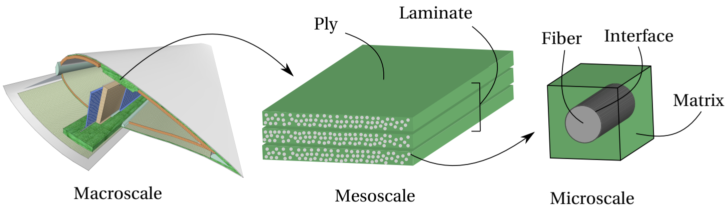
Motivation: Modeling laminated composites

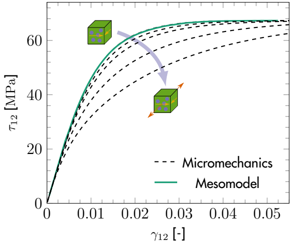
Numerical analysis of composites is an intrinsically multiscale endeavor

- Microscopic failure events $\Rightarrow$ macroscopic failure
- Upscaling: Mesomodel for a single ply $\Rightarrow$ lamination theory $\Rightarrow$ macromodel

Mesoscale models

- Efficiently upscale relevant nonlinear phenomena occurring at the microscale
- However, they can struggle to represent stress states not considered during calibration





[Van der Meer (2016), *Eur J Mech A/Sol*, **60**:58-69]

Back to real constitutive models

Discarding physics has a number of drawbacks:

- Ensuring solution robustness is challenging (negative tangent, spurious localization, etc)
- Non-monotonic strain paths cannot be treated trivially

Physics-based models are robust but not expressive enough:

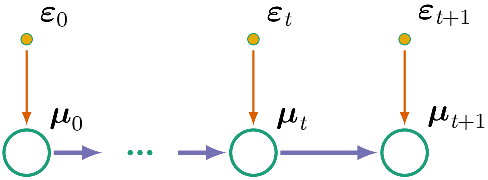
- Constitutive assumptions and fixed pre-calibrated properties

What if material properties could evolve with time?

- Adds freedom to the system, and therefore expressiveness
- Challenge: Properties and internal states cannot be observed. How to infer them?

Inference framework

Increasing flexibility by allowing parameters to evolve in time:



Dynamics:

$$p(\boldsymbol{\mu}_{t+1}) = \mathcal{N}(\boldsymbol{\mu}_{t+1} | \mathcal{A}(\boldsymbol{\mu}_t, \boldsymbol{\varepsilon}_{t+1}), \boldsymbol{\Gamma})$$

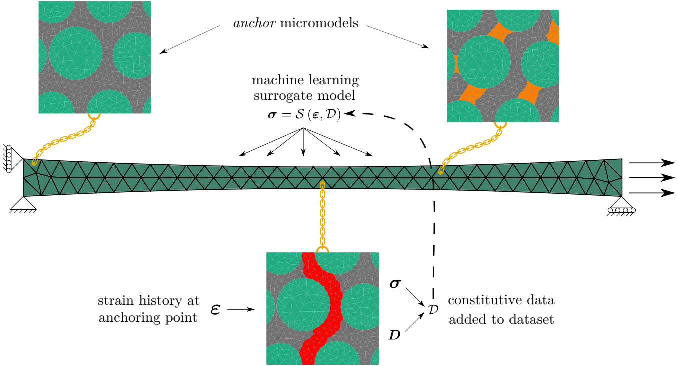
Conditional independence:

$$p(\boldsymbol{\mu}_{t+1} | \boldsymbol{\mu}_t, \boldsymbol{\sigma}_{0:t}) = p(\boldsymbol{\mu}_{t+1} | \boldsymbol{\mu}_t)$$

Back to active learning

Going back to the active learning scheme from before:

- Real plasticity model computing stresses, GP responsible for the yield surface
- History-informed GP input space $\bar{\varepsilon}$ including $\max\limits_{[0:t]}(I_1^{\varepsilon})$ and $\max\limits_{[0:t]}(J_2^{\varepsilon})$



Back to active learning

Predicting elastic unloading:

- Latent yield stress learned from a single anchor at the center of the dogbone